

HARSH SHROFF

SOFTWARE ENGINEER – APPLIED AI & ML SYSTEMS

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PROFESSIONAL SUMMARY

Software Engineer specializing in Applied AI and ML Systems with 4+ years building and shipping production-grade foundation model pipelines, distributed RAG systems, and multi-agent orchestration frameworks. Proven track record translating ambiguous research problems into scalable, measurable engineering solutions — with a growth mindset approach to technical ambiguity. Hands-on experience in LLM fine-tuning (LoRA/QLoRA), responsible AI governance, A/B testing, and MLOps across cloud environments (AWS, GCP). Co-inventor of a patent-pending multimodal AI system under U.S. Army Research Lab funding. AWS ML Certified.

SKILLS

LLM & Foundation Models: RAG Architectures, Fine-Tuning (LoRA/QLoRA/PEFT), RLHF, Multi-Agent Systems, LangGraph, LangChain, LlamaIndex, PydanticAI, vLLM, Prompt Engineering, Gemini, Llama, Claude, GPT, Hugging Face Transformers

Machine Learning & NLP: PyTorch, TensorFlow, Scikit-Learn, Transformer Architectures, DINOv2, ViT, VLMs, Embedding Pipelines, Semantic Search, Intent Detection, A/B Testing, Statistical Modeling

Responsible AI & MLOps: Model Governance, Hallucination Mitigation, Bias Detection, LLM-as-Judge Evaluation, MLflow, Weights & Biases, CI/CD (GitHub Actions), Model Monitoring, Production Observability

Cloud & Infrastructure: AWS (Bedrock, SageMaker, EC2, Lambda, S3, Textract), GCP, Azure (AI Services, OpenAI), Docker, PostgreSQL, pgvector, Pinecone, Redis, Distributed Systems

Software Engineering: Python, C++, FastAPI, SQL, RESTful APIs, Asynchronous Processing, Apache Spark, Kafka, Pandas, Streamlit, Linux/Bash, Git

PROFESSIONAL EXPERIENCE

Applied AI Research Engineer

Mar 2023 – Present

UMBC Center for Real-time Distributed Sensing and Autonomy

- **Architected and deployed a production multi-agent AI system** orchestrating INT4-quantized foundation models (Llama 3.1 via vLLM/Ollama) with event-driven state management and dynamic routing — achieving **sub-8s** end-to-end latency through asynchronous task coordination at the edge.
- **Fine-tuned open-weight LLMs** using LoRA/QLoRA (PEFT) for domain-specific intent classification, establishing LLM-as-Judge evaluation frameworks and automated quality gating to reduce hallucinations prior to production deployment.
- **Built production ML inference pipelines** integrating vision transformers (DINOv2) with language model reasoning via TensorRT and CUDA — cross-functionally collaborating with hardware and software teams to translate research prototypes into shipped capabilities at **50+ FPS** on NVIDIA Jetson.
- **Co-invented a patent-pending multimodal AI system** under the State of Maryland funding, synthesizing RGB, thermal, and audio sensor fusion to improve real-time decision-making systems by **20%**.
- **Established model monitoring and responsible AI governance:** automated drift detection, bias detection, hallucination mitigation, and performance tracking to ensure production reliability and compliance.

AI/ML Engineer – Production AI Systems

Jun 2024 – May 2025

VITG Corp., Halethorpe, MD

- **Architected and shipped a public-facing RAG platform** on EC2 processing 2,500+ municipal regulatory documents with pgvector semantic search and Claude 3.5 (AWS Bedrock) — ensuring high-accuracy retrieval for compliance-critical queries with **100% audit trail coverage**.
- **Engineered a serverless AI automation system** (AWS Lambda + Bedrock) to automate internal screening workflows with dynamic LLM-driven evaluation, reducing manual screening time by **60%** (~25 hrs/week to ~10 hrs/week) while maintaining quality standards.
- **Established enterprise responsible AI governance:** SOC2-compliant logging, automated evaluation metrics, hallucination mitigation strategies, and bias detection pipelines — collaborating with stakeholders to align model outputs with business and regulatory requirements.

- **Designed and ran offline A/B evaluation pipelines** (MLflow) measuring retrieval quality, response accuracy, and P50/P99 latency to iterate on RAG architecture through data-driven experimentation.

Data Scientist – Computer Vision ML

Aug 2023 – Dec 2023

The Conservation Fund, Shepherdstown, WV

- **Deployed an end-to-end production computer vision pipeline** achieving **92% accuracy** in real-time quality assessment on Raspberry Pi edge hardware using YOLOv8 and OpenCV.
- **Co-authored and published peer-reviewed research** in *Elsevier JAFR*, translating deep learning research into commercial impact and enabling automated quality control with production reporting capabilities.

PROJECTS

Multi-Agent Research & Analysis System ([GitHub](#))

LangGraph, Gemini 2.0 Flash, PydanticAI, RAG, LLM-as-Judge, Quality Control Gating, LoRA Fine-Tuning

- Architected an autonomous cyclic AI system orchestrating 11 specialized agents via LangGraph with a self-correcting ReAct loop, Quality Control gating (85/100 threshold), and Planner Agent for optimized search strategies.
- Engineered a production RAG pipeline with chunk-level source grounding, Pydantic-driven structured analysis, and a Citation Verifier to eliminate hallucinations — benchmarked end-to-end via LLM-as-Judge evaluation.

Bio-Oracle: Neuro-Symbolic Agentic AI ([GitHub](#))

PydanticAI, Gemini, Cellpose, Docker, Structured Reasoning, Verifiable AI

- Developed a Neuro-Symbolic Agent for high-throughput phenotypic drug discovery screening on BBBC021 — combining deep learning segmentation with structured LLM reasoning for verifiable, auditable outputs.
- Architected using PydanticAI for schema-enforced structured outputs, packaged in a production-ready Docker container with reproducible local inference.

Multi-Modal Recommendation Engine ([GitHub](#))

Computer Vision, LLM Personalization, A/B Testing, PostgreSQL, Candidate Ranking

- Engineered a hybrid recommendation system combining computer vision and LLM-based personalization with scalable candidate generation and ranking — A/B tested for continuous improvement, reducing user bounce rate by **20%** and increasing engagement by **40%**.

Silicon Oracle: Asynchronous LLM Intelligence Engine ([GitHub](#))

Flask, PostgreSQL, Gemini 2.0 Flash, APScheduler, Multi-Tenant Architecture, BYOK

- Engineered a multi-tenant async ML pipeline (APScheduler + PostgreSQL) orchestrating parallelized Gemini 2.0 Flash LLM inference across concurrent users with a proprietary 15-factor scoring system and Fernet-encrypted BYOK cryptographic architecture.

CHARLIE v2.0: Local Multi-Agent AI Assistant ([GitLab](#))

Python, Llama 3.1 (GGUF), Ollama, CUDA, TensorRT, DINOv2, vLLM

- Architected a Jarvis-inspired local AI assistant integrating quantized Llama 3.1 inference, voice interaction, and multimodal perception into a unified multi-agent system for contextual task execution.
- Built a privacy-focused offline inference stack using Ollama/vLLM with optimized GPU acceleration, dynamic tool routing, and modular agent communication for scalable AI-assisted workflows.
- Engineered an asynchronous agent orchestration pipeline combining wake-word activation, speech-to-text, LLM reasoning, text-to-speech, and CUDA-accelerated computer vision (DINOv2/TensorRT) for real-time scene understanding and adaptive decision-making.

EDUCATION

Master of Science, Data Science

Aug 2022 – May 2024

University of Maryland Baltimore County (UMBC) — GPA: 3.8/4.0

CERTIFICATIONS & PUBLICATIONS

Professional Certifications

- **AWS Machine Learning Associate** – [View Credential](#) – Active through Apr 2028
- **NVIDIA Deep Learning Institute** – Transformer-Based NLP Applications; GPU-Accelerated Computing

Publications & Research

- **Ranjan, R., Shroff, H., et al.** (2024). “FilletCam AI: Precision color profiling using deep learning.” *Journal of Agriculture and Food Research*. DOI: [10.1016/j.jafr.2024.101461](#)
- **Trivedi, K., Shroff, H.** (2021). “Mosquito identification using ML on embedded systems.” *IEEE ITU Kaleidoscope* (Ranked 1st Internationally). DOI: [10.23919/ITUK53220.2021.9662116](#)